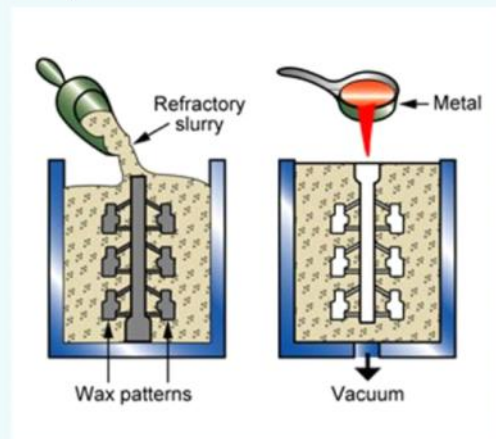


Week 5 - Quiz

21 May 2025 11:44

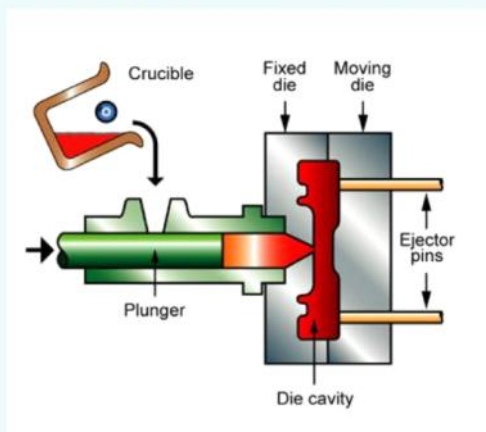
Which process is this?



Select one:

- ☐ a. Gravity Die Casting → No die in image
- ☒ b. Investment Casting
- ☐ c. Die Casting → No die
- ☐ d. Sand Casting → No sand

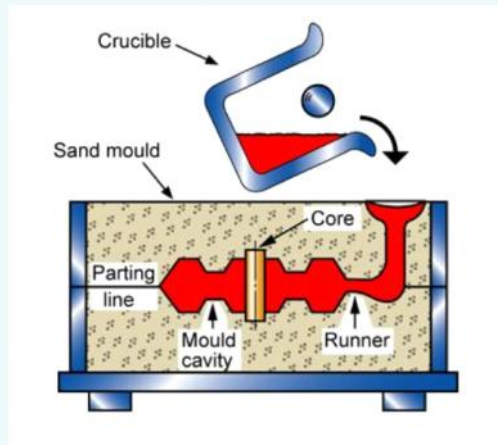
Which process is this?



Select one:

- ☐ a. Sand Casting → No sand
- ☐ b. Investment Casting → Die present, so cannot be this
- ☐ c. Gravity Die Casting → No gravity
- ☒ d. High Pressure Die Casting

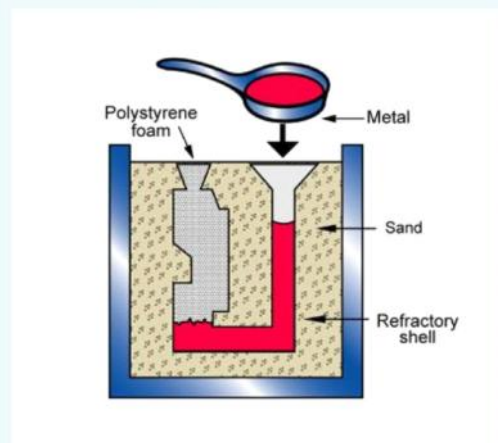
Which process is this?



Select one:

- ☐ a. Die Casting → No die
- ☐ b. Investment Casting → Sand present, so cannot be.
- ☐ c. Uphill teaming → ?
- ☒ d. Sand Casting

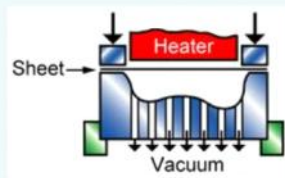
Which process is this?



Select one:

- ☐ a. Low Pressure Sand Casting → No pressure present
- ☒ b. Evaporative Pattern Sand Casting
- ☐ c. Investment Casting → Sand present.
- ☐ d. Gravity Die Casting → No die

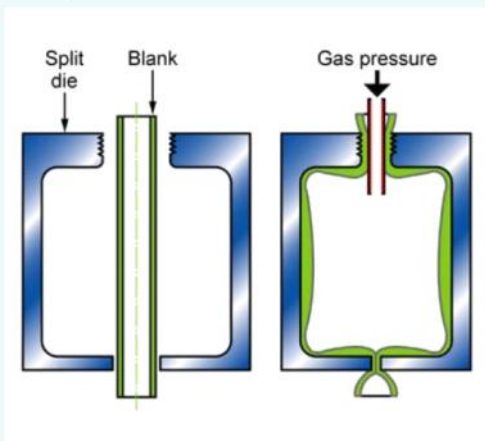
Which process is this?



Select one:

- ☒ a. Vacuum Forming → Vacuum present
- ☐ b. Drape Forming
- ☐ c. Pressure Forming
- ☐ d. Compression Moulding

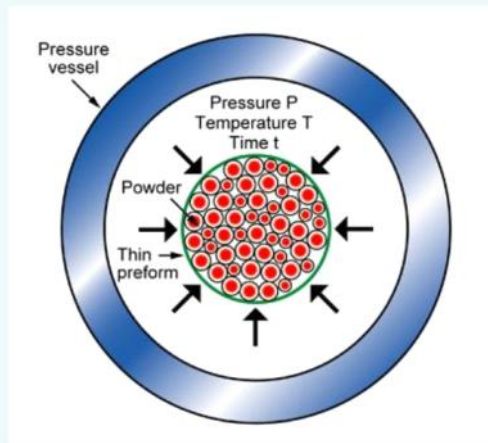
Which process is this?



Select one:

- ☐ a. Injection Moulding
- ☐ b. Rotational Moulding
- ☐ c. Extrusion
- ☒ d. Blow Moulding → Gas used to blow blank into mould.

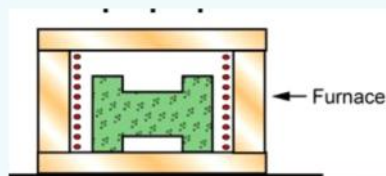
Which process is this?



Select one:

- ☒ a. Hot Isostatic Pressing
- ☐ b. Tape Casting
- ☐ c. Hot Pressing
- ☐ d. Sintering

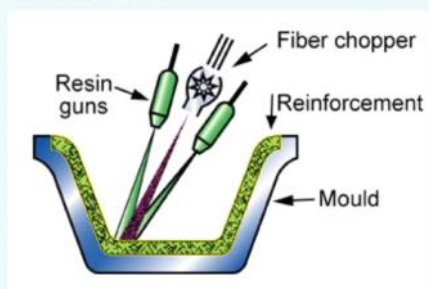
Which process is this?



Select one:

- ☒ a. Sintering
- ☐ b. Hot Pressing
- ☐ c. Tape Casting
- ☐ d. Hot Isostatic Pressing

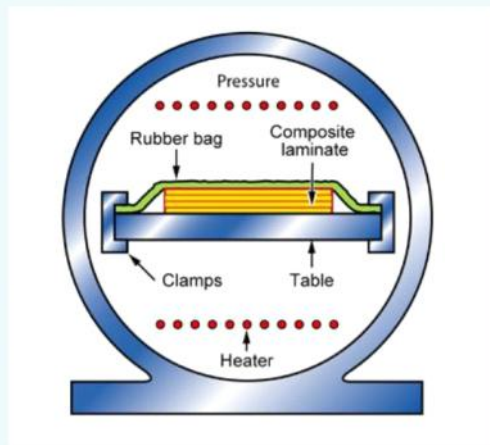
Which process is this?



Select one:

- ☐ a. Sheet Moulding Compound Moulding
- ☐ b. Dough Moulding Compound Moulding
- ☐ c. Filament Winding
- ☒ d. Spray Lay Up

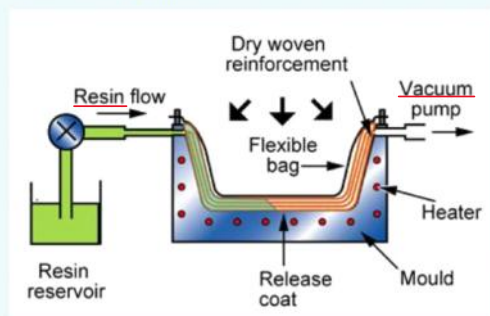
Which process is this?



Select one:

- ☐ a. Resin Transfer Moulding
- ☐ b. Hand Lay Up
- ☒ c. Autoclave Moulding
- ☐ d. Pultrusion

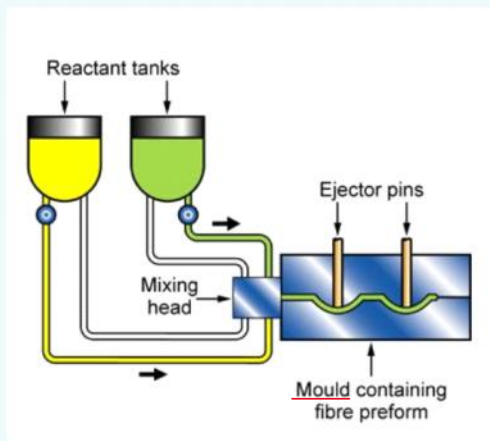
Which process is this?



Select one:

- ☒ a. Vacuum Assisted Resin Transfer Moulding
- ☐ b. Hand Lay Up
- ☐ c. Pultrusion
- ☐ d. Autoclave Moulding

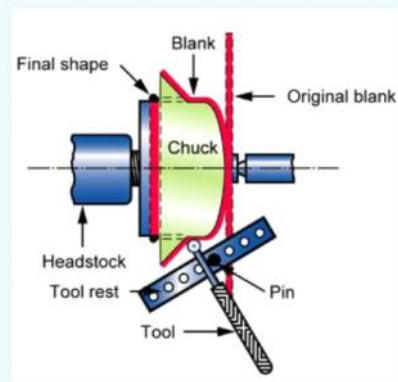
Which process is this?



Select one:

- ☐ a. Autoclave Moulding
- ☒ b. Resin Transfer Moulding
- ☐ c. Pultrusion
- ☐ d. Hand Lay Up

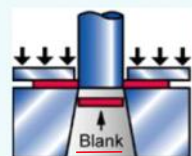
Which process is this?



Select one:

- ☐ a. Sheet Stamping
- ☒ b. Spinning
- ☐ c. Forging
- ☐ d. Sheet Forming

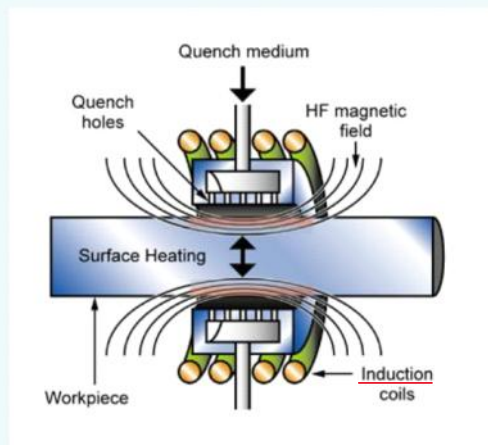
Which process is this?



Select one:

- ☐ a. Bending
- ☐ b. Deep Drawing
- ☐ c. Stretching
- ☒ d. Blanking

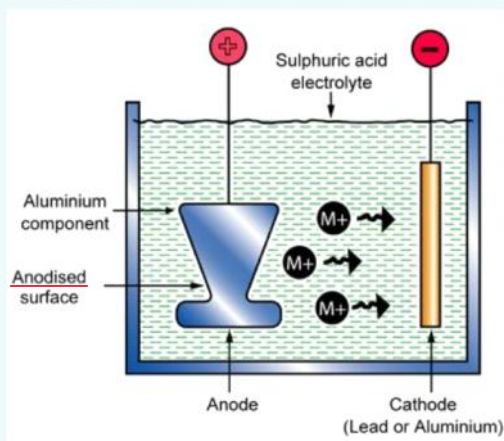
Which process is this?



Select one:

- ☐ a. Laser Hardening
- ☐ b. Flame Hardening
- ☐ c. Nitriding
- ☒ d. Induction Hardening

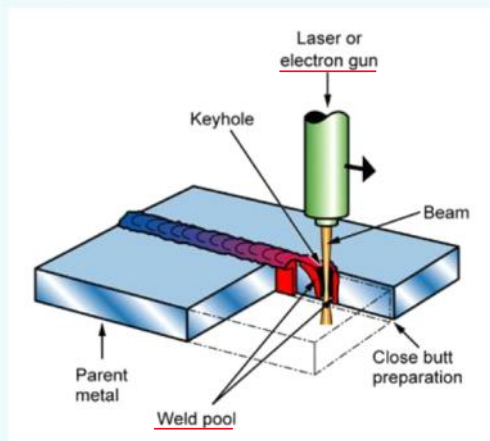
Which process is this?



Select one:

- ☐ a. Galvanising
- ☒ b. Anodising
- ☐ c. Hot Dip Coating
- ☐ d. Electroplating

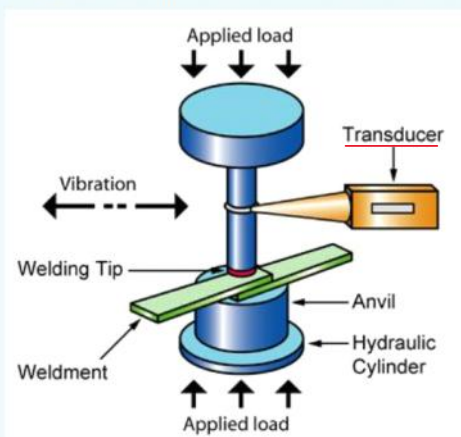
Which process is this?



Select one:

- ☐ a. Oxyacetylene welding
- ☐ b. Brazing
- ☐ c. Ultrasonic welding
- ☐ d. Gas tungsten arc welding
- ☐ e. Friction Stir Welding
- ☒ f. Electron beam welding
- ☐ g. Gas metal arc welding

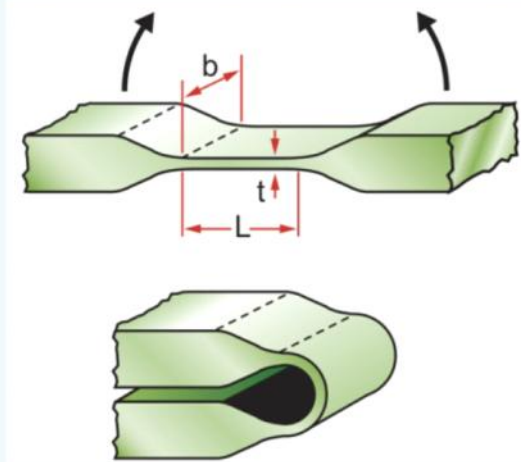
Which process is this?



Select one:

- ☐ a. Gas tungsten arc welding
- ☐ b. Gas metal arc welding
- ☐ c. Oxyacetylene welding
- ☒ d. Ultrasonic welding
- ☐ e. Friction Stir Welding
- ☐ f. Power beam welding
- ☐ g. Brazing

Lets consider the hinge for the lid of a box such as a DVD case. Then the hinge is a thin ligament that flexes elastically as the lid is closed. Then the hinge is formed along one side of the opening. The aim is to maximise the movement (or the angle over which it can be rotated) without resulting in any permanent damage. The best material is the one (for a given ligament dimension) that can be bent to the smallest radius without yielding or breaking. To reduce costs to the lowest level presume that the hinge is made from the same material as the box and the lid and that they are made in a single manufacturing process together.

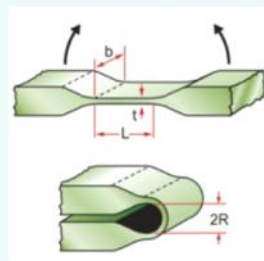


Having read this carefully now identify what the correct set of constraints, free variables and objectives are for this design.

Select one:

- ☐ Constraints - Material choice
Free Variables - No failure
Objective - Elastic flexure
- ☒ Constraints - No failure
Free Variables - Material choice
Objective - Elastic flexure
- ☐ Constraints - Elastic flexure
Free Variables - No failure
Objective - Material choice
- ☐ Constraints - No failure
Free Variables - Elastic flexure
Objective - Material choice

If the objective is to minimise the radius that the hinge can bend to without breaking or yielding. We need the radius to be expressed in terms of other parameters. So which of the following tells us the tensile surface strain on the outside of the hinge, which is of thickness t bent to a radius of R ?



Select one:

- ☐ $\epsilon = t/R$
- ☒ $\epsilon = t/2R$
- ☐ $\epsilon = bt/2LR$
- ☐ $\epsilon = t/L$

$$\epsilon = \frac{\Delta L}{L} = \frac{t}{2R}$$

If the hinge is still behaving elastically, then what would the surface stress be?

Select one:

- ☐ $\sigma = \frac{2ER}{t}$
- ☒ $\sigma = \frac{Et}{2R}$
- ☐ $\sigma = \frac{t}{2ER}$
- ☐ $\sigma = \frac{2R}{Et}$

$$E = \frac{\sigma}{\epsilon} \Rightarrow \sigma = E\epsilon = \frac{Et}{2R}$$

The minimum radius for a given material is achieved when the material is at its yield strength. What is the radius at that point?

Select one:

- ☐ $R = \frac{\sigma_f}{Et}$
- ☒ $R = \frac{Et}{2\sigma_f}$
- ☐ $R = \frac{2\sigma_f}{E}$
- ☐ $R = \frac{E}{t\sigma_f}$

$$\sigma_f = \frac{Et}{2R}$$

$$R = \frac{Et}{2\sigma_f}$$

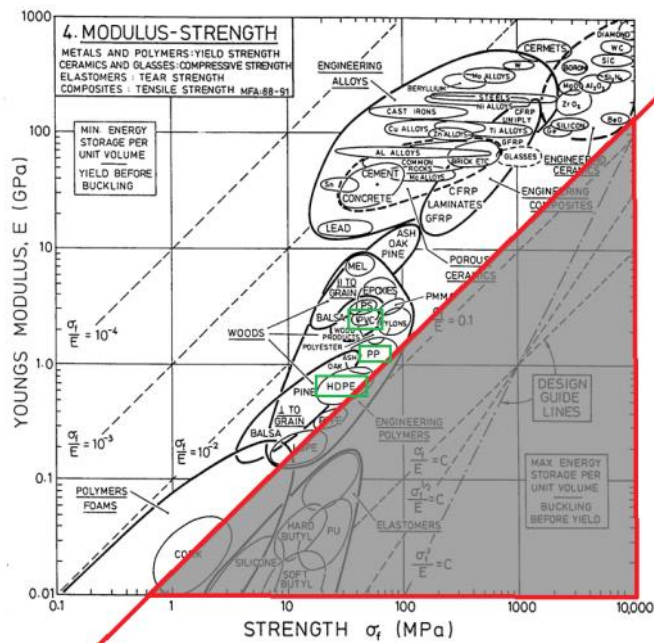
Therefore the material that can be bent to the smallest radius is the one that has the greatest value of the materials index.

Select one:

- ☒ $M_1 = \frac{\sigma_f}{E}$
- ☐ $M_1 = \frac{\sigma_f}{E^2}$
- ☐ $M_1 = \frac{E}{\sigma_f}$
- ☐ $M_1 = \frac{E\rho}{\sigma_f}$

$$\downarrow R = \left(\frac{t}{2}\right) \left(\frac{E}{\sigma_f}\right) \downarrow = \text{Minimise}$$

$$\text{Greatest value} = \text{Maximise} = \frac{\sigma_f}{E}$$



Match the Following Products to the Method of Manufacture



Choose...

- High Pressure Die Casting
- Low Pressure Die Casting
- Evaporative Pattern Sand casting**
- Low Pressure Sand Casting
- Gravity Die Casting
- Ceramic Shell and Evaporative Mould Casting



Choose...

- High Pressure Die Casting
- Low Pressure Die Casting
- Evaporative Pattern Sand casting
- Low Pressure Sand Casting
- Gravity Die Casting
- Ceramic Shell and Evaporative Mould Casting**



Choose...

- High Pressure Die Casting
- Low Pressure Die Casting**
- Evaporative Pattern Sand casting
- Low Pressure Sand Casting
- Gravity Die Casting
- Ceramic Shell and Evaporative Mould Casting



Choose...

- High Pressure Die Casting
- Low Pressure Die Casting
- Evaporative Pattern Sand casting
- Low Pressure Sand Casting
- Gravity Die Casting**
- Ceramic Shell and Evaporative Mould Casting

Match the Following Products to the Method of Manufacture



Choose...

- Extrusion
- Injection Moulding
- Expanded Foam Moulding
- Blow Moulding
- Rotational Moulding**
- Thermoforming



Choose...

- Extrusion
- Injection Moulding
- Expanded Foam Moulding**
- Blow Moulding
- Rotational Moulding
- Thermoforming



Choose...

- Extrusion**
- Injection Moulding
- Expanded Foam Moulding



Choose...

- Injection Moulding
- Expanded Foam Moulding
- Blow Moulding
- Rotational Moulding
- Thermoforming



Choose...

- Extrusion
- Injection Moulding
- Expanded Foam Moulding
- Blow Moulding
- Rotational Moulding
- Thermoforming